

Project Proposal — Fitness Tracking Application

Overview

Our project is a fitness tracking application designed to help users keep track of their workouts and fitness progress in one place. Users will be able to record their exercises, sets, repetitions, weight, and other workout information. The application will organize this information so users can easily look back at their previous workouts and monitor their progress over time.

Problem

Keeping track of workouts can be difficult when information is written down in different places or not recorded at all. Our application will provide users with a simple way to record their workouts and view their progress over time without having to manually organize their fitness information.

Goals

The main goal of this project is to create an easy-to-use fitness tracking application that allows users to:

- Create and manage their workout information.
- Record exercises, sets, repetitions, and weight.
- View their previous workouts.
- Track their progress over time.
- Organize workouts into different routines or workout sessions.

Target Users

The application is primarily intended for people who regularly exercise and want an easy way to track their workouts and progress. It can be used by beginners as well as more experienced gym users.

Example users:

Marcus, 24 — consistent lifter

Trains four days a week on a push/pull/legs split. Currently tracks lifts in his phone's notes app, which makes it tedious to find what he did the last time he trained a given movement. Needs fast entry during a session and a quick way to see whether a lift is trending up. Constraint: enters data one-handed, between sets, on a phone.

Priya, 19 — new to the gym

Started lifting two months ago and follows a beginner routine she found online. Isn't sure whether she's actually progressing and forgets what weight she used last week. Needs a saved routine she can follow and a simple history she can look back at. Constraint: unfamiliar with lifting terminology, so labels need to be plain.

Tech stack

The application will be built as a React single-page frontend communicating over a REST API with a Node.js/Express backend, backed by a PostgreSQL database

We chose this stack because both team members already work in JavaScript and SQL, which keeps the semester focused on software-development practice rather than learning new languages, and because the relational structure of the data — users owning workouts, workouts containing exercises and sets — maps naturally onto a relational database.

Planned Features

Some of the initial features we plan to implement include:

- User accounts and profiles.
- Workout logging.
- Exercise tracking.
- Workout history.
- Personal workout routines.
- Progress tracking.
- A dashboard showing basic fitness statistics.

Additional features may be added as the project develops based on the team's available time and priorities.

2. GITHUB LINK - <https://github.com/ansel-stinnett/reprog>

3. Initial readme and backlog

Readme -

<https://github.com/ansel-stinnett/reprog/blob/main/README.md>

Backlog -

<https://github.com/ansel-stinnett/relog/blob/main/BACKLOG.md>

4. Initial Architecture

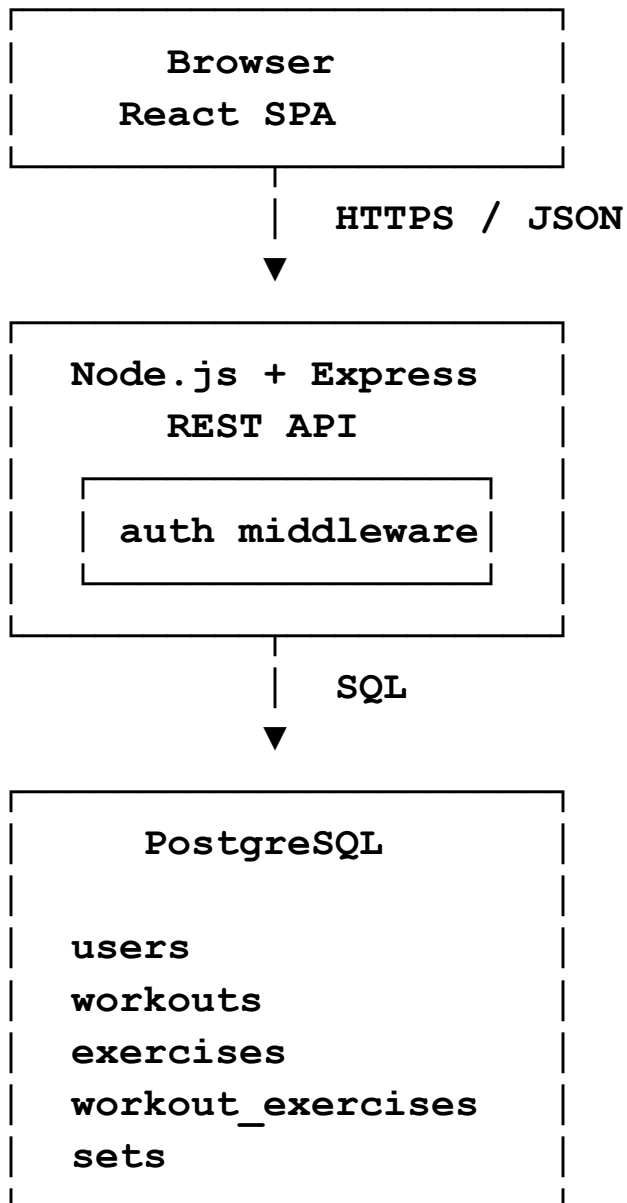


Figure 1: System context diagram

5. Team contributions

Nhan Trinh: Nhan will contribute to the development of the fitness tracking application's backend and database. He will also assist with testing and debugging the application.

Ansel Stinnett: Ansel will design the REST API contract and implement authentication and the workout-logging features across the frontend and assist with backend development. He will also set up the repository, branching workflow, and code review process, and share responsibility for testing and documentation.